

Answers

1 → D

2 → C

3 → ~~D~~ A

4 → D

5 → A

6 → B

7 → B

8 → D

9 → D

10 → B

11 → B

12 → C

13 → B

14 → C

15 → B

16 → C

17 → A

18 → C

19 → B

20 → B

21. Supervised Learning is used for prediction based on Labelled training data. The algorithm learn a function from the model input variables (feature) to the output variables (Target).

25. List two major application of data science :

- Healthcare
- bank

26. Distinguish between the term structured and semi-structured data :

- Structured data : is the data with high organization.

Example : CSV file , spread sheet file

- Semi-structured data : is the data with some organization .

Example : HTML file , XML file .

23. Regression analysis is an analysis that used for numerous problem. It predicts the value of a variable based on the value of the other variables .

34. Comment the performance of the classifiers :

Represent them as A, B and C in order :

- Classifier A is the best case : the hyperplanes separate the data into two classes correctly. The distance between the hyperplanes also great because they are far enough from each other.
- Classifier B is the worst case : the hyperplanes does not separate the data/object well since there are some remaining data in between the hyperplanes.
- Classifier C is good but not the best : the hyperplanes separate the data/object well but the distance between hyperplanes is too short.

30. The continuous attributes are discretized by convert them using one-hot encoding or use dummy technique.

29. Data normalization is required in KNN because data normalization is a process of scaling the value of variable is the value between 0 and 1. data normalization allow the comparison between the data that has difference unit or scale in KNN.

27.

31. K-mean algorithm is an iterative algorithm that tries to partition the dataset into K pre-defined distinct non-overlapping (cluster) where each data point belong to only one group.

32. Name few hyperparameters :

- Linear kernel
- polynomial kernel
- Radial kernel.

28. Data pre-processing is a process that take raw data and transform it into a format that can be analyze and understood by a computer.

24. two challenges of machine learning :

- Data preparation : real life data is messy, need a lot of cleaning.
- Data security and privacy